

Models	DM	CD	WK
Google ngram	0.312 ₀	0.222 ₀	0.442 ₀
BERT-b	0.569 ₂₁	0.539 ₂₀	0.770 ₁
BERT-I	0.569 ₃₂	0.558 ₂₁	0.738 ₃₂
RoBERTa-b	0.544 ₁₄	0.527 ₁₉	0.689 ₁₄
RoBERTa-I	0.524 ₂₉	0.585 ₁₇	0.754 ₄
Falcon	0.452 ₁₁	0.567 ₁₈	0.623 ₇
GPT-4	0.666	0.739	0.852
Flan-T5-xl	0.684 ₁₇	0.621 ₁₇	0.869 ₁₆
Flan-T5-xxl	0.633 ₁₆	0.655 ₁₇	0.787 ₁₆

Table 5: Indirect scalar intensity ranking table. Subscripted numbers are the best-performing template number. The best results across models are in bold.

We found that models all encode rich information about adjective intensity by significantly outperforming the baseline. For BERT and RoBERTa, we again find that direct and indirect probing show similar relative performance, with absolute results for indirect probing lower than those for direct probing.

GPT-4 dominates again in this task as it largely outperforms all models in one dataset and has competitive performance with Flan-T5-xl in the other two. The scaling law does not explain the pattern of results; across different families, Falcon is much larger than BERT and RoBERTa models but performs much worse. Even within encoder models, RoBERTa also fails to outperform BERT.

4. Probing Scalar Diversity Pragmatics

Given that LLMs have different levels of lexical-semantic knowledge about SAs, we can now ask whether these different levels correlate with the LLMs’ ability to draw pragmatic inferences. This section assesses one particular aspect of this question, namely whether models can reason about the scalar diversity of SAs.

4.1. Scalar Diversity Probing Models

In this section, we describe how we use naturalistic settings to probe LLMs to answer questions about scalar diversity. We only use instruction-tuned models in the previous sections, namely Falcon, GPT-4, Flan-T5-xl/xxl so that they can properly follow instructions.

4.2. Scalar Diversity Probing Dataset

We use all SI instances triggered by SAs from **PVT** (Pankratz and van Tiel, 2021), **GZ** (Gotzner et al., 2018), and **RX** (Ronai and Xiang, 2022), providing a total of 152 instances. About 40 human participants answer yes or no to each prompt such as

‘Mary: The problem is *hard*. Would you conclude from this that Mary thinks the problem is not *unsolvable*?’ Since 40 annotators may not represent very fine-grained human judgments, we convert these datasets to binary classifications: we force models to answer *yes* or *no* given a question similar to the approach used in Ruis et al. (2024) (see the next subsection for an illustration), and consider the answers to be *yes* for instances to which at least half of the participants infer SIs, otherwise *no*.

An overview of these datasets can be found in Table 6.

Dataset	Total instance	Yes	No
PVT	50	13	37
GZ	70	19	51
RX	32	5	27

Table 6: Overview of total instances and answer counts in scalar diversity datasets.

The above datasets are relatively small. However, there are no existing large-scale datasets for scalar diversity triggered by SAs. By using three such datasets, we hope to provide a reasonable evaluation of the models’ capabilities.

4.3. Scalar Diversity Probing Method

We note that some models may have inherent preferences for *yes* or *no* (e.g. more likely to answer *yes* to neutral prompts or vice versa). To debias them for such preferences, we adopt the following strategies. First, we feed the full prompt to models and record the probability for *yes* and *no* as the immediate token following the prompt and denote them as sy and sn , respectively. For models other than GPT-4, we apply three strategies to retrieve their best performance: (i) sy , (ii) weighted probability for sy (denoted as wy) and (iii) calibrated probability of wy (denoted as cy). For each strategy, we consider the models’ answer to be *yes* if the probability is at least 0.5, *no* otherwise.

The value of wy is calculated as Equation 3:

$$wy = \frac{sy}{sy + sn} \quad (3)$$

cy is the calibrated probability of wy , which we calibrate with the probability distribution after softmax as Zhao et al. (2021) in practice. We use three neutral fillers [N/A], empty string, and [MASK] in positions of scalar claims, which give us six different neutral contexts for each template. An example is provided below, where the black part is prompt and the blue part is the answer option.

Question: Imagine that your friend Mary says, "[MASK]"
Would you conclude from this

that Mary thinks [N/A]?
Only answer yes or no.
Answer: yes/no

Given these neutral prompts, we average the probabilities of w_y in each prompt and calculate a calibration weight matrix W which calibrates the averaged w_y to be 0.5 (i.e., neutralize it). We then use this matrix to calibrate w_y in non-neutral prompts to derive c_y .

4.4. Scalar Diversity Experiment and Results

Because the datasets are unbalanced, we use macro F1 as the evaluation metric. We use a logistic regression (LR) model with string-based and concept-based surprisal features (Hu et al., 2023) as a baseline.¹¹ When making SI predictions in one dataset, we use either of the remaining datasets or both of them to train an LR model. We report the best performance and strategies for all models in Table 7.¹²

Model	RX	GZ	PVT	Avg
LR	0.726 _{RX}	0.713 _{RX+PVT}	0.688 _{GZ}	0.709
Falcon	0.458 _{sy}	0.534 _{cy}	0.578 _{cy}	0.464 _{cy}
Flan-T5-xl	0.835 _{cy}	0.757 _{cy}	0.746_{cy}	0.779 _{cy}
Flan-T5-xxl	0.897_{sy}	0.867_{cy}	0.726 _{cy}	0.816_{cy}
GPT-4	0.759	0.598	0.729	0.695

Table 7: Macro-F1 scores for all models across datasets. Subscripted letters for the baseline model (LR) refer to its training dataset. The average result for LR is taken over the best per-dataset results as a universal training strategy for all datasets does not exist. Subscripted letters for non-baseline models refer to strategies used. The best results across models are in bold.

Generally, we find that decoder models have unsatisfying performance in this task. For instance, Falcon performs below baseline, and GPT-4 only performs on par with the baseline and underperforms Flan-T5 models. Flan-T5-xxl shows the best results, despite having much space to reach ceiling performance. This result is slightly different from generalized implicature probing results from Ruis et al. (2024), which reports that GPT-4 outperforms Flan-T5 models.¹³

¹¹We drop 4 datapoints in **GZ** in the baseline as they do not have the surprisal information.

¹²See full results for non-baseline models (except GPT-4, which we do not have access to its probability distribution by the time we conduct this experiment) in Appendix 11.8.

¹³We hypothesize this result could be for two reasons. First, their dataset contains very few direct scalar adjective comparisons. Second, we report zero-shot results instead of few-shot and chain-of-thought results, although

5. A Critical Appraisal of Results

5.1. ‘Bad’ Prompts, Good Results

Recall the following surprising result from §3.3: using simply concatenated adjective strings as inputs – which are unnatural – yields better direct scalar intensity ranking results than natural contexts with single adjectives (G&A).

From the perspective of human language processing, recent experimental linguistic works show that humans derive more robust scalar contrasts when both strong and weak terms appear salient in the same context (Ronai and Xiang, 2023a). It seems reasonable to assume that on different days, a speaker may describe a movie as *good* or *awesome*, but actually have the same degree estimate mentally (e.g. *the movie is a 7/10*) despite these terms having different semantic intensities in general. However, when using these contrastive adjectives in the same context, it is less likely that they have the same mental degree estimates. LLMs may work similarly when computing adjective intensities.

However, we also note that previous works suggest that LLMs tend to memorize information from training (Zhong et al., 2021) while our prompts are not natural at all (e.g. *good awesome wonderful great* does not exist in Google Ngram corpus). It is unclear why the ‘bad’ prompting method performs better than natural G&A templates. We hypothesize that attention only picks up salient discourse elements (i.e. the comparative adjectives), which is visualized in Figure 3. Natural contexts containing these adjectives, which might appear in training, may not differ as much from our prompt when computing SA representations.

5.2. Indirect Probing: Lower-bounded Absolutely, Faithful Relatively

Indirect probing shows unsatisfying absolute results compared to direct probing in §3. This finding aligns with Petroni et al. (2019), albeit the relative results remain unchanged. There are several possible explanations. First, although LLMs encode SA lexical semantics in word embeddings, they cannot reason about them in complex linguistic constructions. Second, direct probing in our study utilizes the best layer-wise information. The indirect method may not be as good for comprehensively selecting useful information across model layers. Third, although we have many prompts to get the best performance out of models, they could nevertheless be biased by the training corpus distribution if these prompts are not frequent enough (Zhong et al., 2021).

Ruis et al. (2024) notes that these techniques do not help GPT-4 performance.

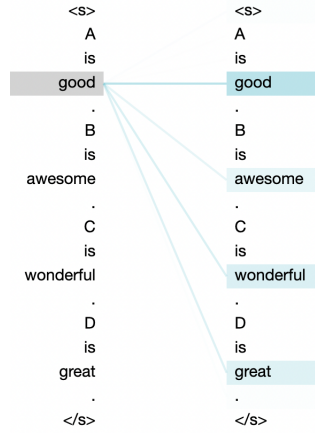


Figure 3: Attention visualization by Bertviz (Vig, 2019). Attention head 10 in the last layer of RoBERTa-b picks up *good*, *great*, *wonderful*, *awesome* when computing *good* in the context of ‘A is good. B is awesome. C is wonderful. D is great.’

5.3. Scaling Does Not Explain Performance

In the lexical-semantic task (§3), the scaling law tends to hold. However, it does not explain all the patterns in the results: RoBERTa underperforms smaller BERT, Falcon underperforms smaller encoder models, and Flan-T5-xl. Lorge and Pierrehumbert (2023) also reports similar observations that BERT has better representations of scalar adverbs than RoBERTa. We hypothesize that this can be explained by training objectives. First, models that model sentence relations in pre-training tend to learn fine-grained lexical semantics easier than those without: BERT features next-sentence prediction, and Flan-T5 models sentence relations in the encoders and decoders, which forces them to learn semantics contrastively, which aligns with the findings in Merrill et al. (2024). Moreover, the fact that Falcon underperforms RoBERTa, which also lacks sentence relation training, can be explained by that the decoders only model left-hand context, which constrains them from reasoning about useful right-hand contexts. However, it is nevertheless surprising that Flan-T5-xxl underperforms the smaller Flan-T5-xl. We leave this question for future investigations.

5.4. Good Lexical Semantics Does Not Entail Good Pragmatics

In scalar diversity reasoning (section 4), GPT-4 performs worse than the much smaller Flan-T5 models, even though it is the best-performing model in the lexical-semantic task. We find that GPT-4 predicts *no* for at least 90% of the instances. Instead of forcing a *yes* or *no* answer in section 4, we try free generation here to gain further insights. Figure 4

displays one instance where humans are confident in saying *yes*, but GPT4 says *no*. We see that the model is quite conservative in making statements about implicit status.¹⁴

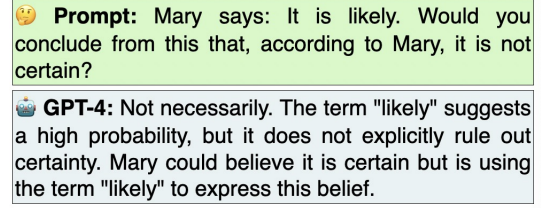


Figure 4: Free generation results for GPT-4 using a prompt from GZ without forcing *yes* or *no* answers.

Considering linguistic capabilities, we may not always expect models that are good at one task to be good at another. Weissweiler et al. (2022) also note that better syntax does not always mean better semantics. Specifically, the gap between semantic and pragmatic capabilities in our work can be explained by the fact that a weaker scalar alternative does not always give rise to the negation of the stronger one. Humans consider various factors in addition to lexical semantics such as the discourse context, the frequency of the implicature (Pankratz and van Tiel, 2021), politeness (Gotzner and Mazzarella, 2021), and other social factors. Even more confoundingly, humans also need to consider the possibility that conversational implicatures might be canceled in the subsequent discourse (Potts, 2015). Due to these complicated concerns, models need to both understand SA lexical semantics and reason pragmatically about various contextual factors to derive implicatures. Thus, good semantic models may not be good pragmatic models in reasoning about communicative intentions.

6. Conclusion

This paper has shown that LLMs of different sizes and architectures encode rich information about SA lexical semantics. Moreover, we have shown that good lexical-semantic knowledge of SAs does not always give rise to good performance in the pragmatic reasoning task about their scalar diversity. Finally, we leveraged both linguistic intuitions and model training objectives to provide an analysis for our probing results.

¹⁴Although conversational implicatures can be canceled (Grice, 1975) for discussion), we wish to emphasize that the paper investigates the degree to which LLM behavior matches that of an independently collected, normed, dataset. The possibility of cancellation is less evident to the experimental participants in the dataset collection.